Unit 202 — Four-Asset Pilot Review

Package: CC-11.9 · Generated: 2026-08-25T17:58:59.284Z

ALL FOUR PASS — AUTOMATIC GO INTO FULL PRODUCTION

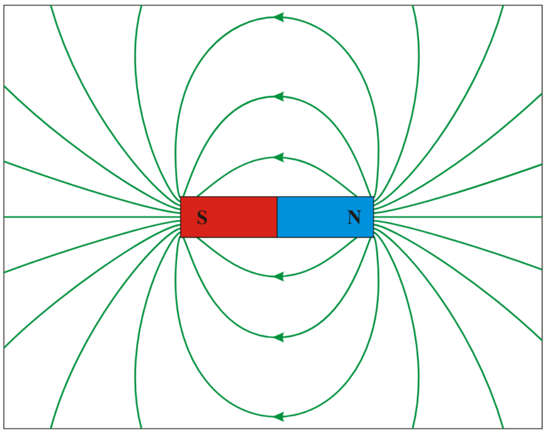
Assets attempted	4
PASS	4
RETRY (mid-pipeline, superseded by a later attempt)	0
HUMAN_REVIEW_REQUIRED	0
Not completed	0

Bar magnet and magnetic field

unit202.magnet.field

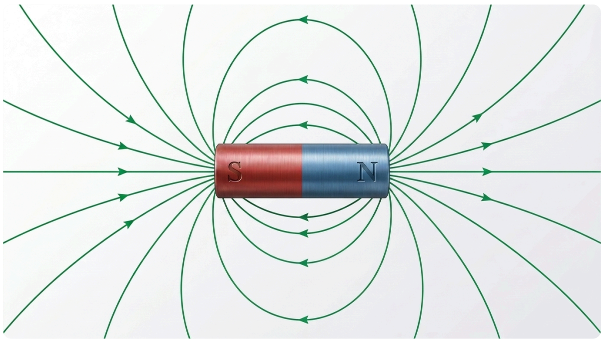
Curriculum context	LO5 — lesson.magnetism.fundamentals
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a bar magnet with its external magnetic field pattern, N to S, as the basis for flux/flux-density teaching.

REFERENCE (AS SENT TO GEMINI)



Wikimedia Commons — DipolMagnet.svg

FINAL ALP ARTWORK (ATTEMPT 4)



Model: gemini-3.1-flash-image · Generated: 2026-08-25T17:37:57.971Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
meaningful field-line geometry	PASS	Full nested-loop topology present and unchanged from v3, minus one non-topological rendering artefact (see below).
external field direction runs N to S	PASS	Re-verified via zoomed crops of both horizontal-ray arrowheads: the N-side (right) ray points outward, away from the magnet (field exiting N); the S-side (left) ray points inward, toward the magnet (field entering S). The nested arc loops all point toward S at their peaks, consistent with the same N-to-S circulation. All three line families agree with each other -- no self-contradiction.
field-line density used meaningfully where density is taught	PASS	Not applicable to this state (density_comparison=false).

Prohibited change	Result	Note
do not draw field lines that reverse direction or cross incorrectly	PASS	No reversal or incorrect crossing found on re-inspection.

Overall findings	PASS. This asset's technical geometry (field-line topology and direction) was correct as of v3; the only real defect was a cosmetic rendering artefact (a faint duplicate ghost arc), now removed via one narrow cleanup pass that left every technical element unchanged (independently re-verified via zoomed pixel comparison, not merely assumed). unit202.magnet.field is production-candidate-ready pending Product Owner review.

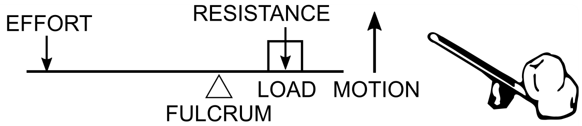
Master path	`reports/instructional-visuals/premium-artwork/proof/unit202.magnet.field/unit202.magnet.field-master-v4.png`

Lever — Class I

unit202.levers.class-1

Curriculum context	LO3 — lesson.foundation.physics.simple-machines
Pedagogical role	CONFIGURATION
Production class	HYBRID
Purpose	Show a Class I lever arrangement (fulcrum between effort and load) so a learner can recognise it from the arrangement itself.

REFERENCE (AS SENT TO GEMINI)



Pearson Scott Foresman — Lever (PSF).svg

FINAL ALP ARTWORK (ATTEMPT 3)

Model: gemini-3.1-flash-image · Generated: 2026-08-24T23:20:43.588Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
fulcrum positioned between the effort point and the load point	PASS	Single lever bar with the fulcrum wedge clearly positioned between the EFFORT end (left, with a hand-grip loop) and the LOAD end (right, with a cube load). Correct Class I ordering.

Prohibited change	Result	Note
do not blend this with the Class II or Class III arrangement	PASS	Exactly one lever diagram is present. No trace of the Class II or Class III sibling arrangements -- the prepared/cropped reference (isolating only the top subfigure) resolved the composite-reference defect that caused attempts 1-2 (CC-11.8) to fail on this exact point.

Overall findings	Strong PASS on first attempt of this production run. The composite-reference crop (isolating the Class I subfigure before sending to Gemini) fully resolved the defect that blocked this asset in the earlier CC-11.8 proof (which had sent the full 3-class composite and got a blended result). No immutable fact violated, no prohibited change made, required labels present and correct.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.levers.class-1\unit202.levers.class-1-master-v3.png`

Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF

unit202.generator.rotating-loop.horizontal

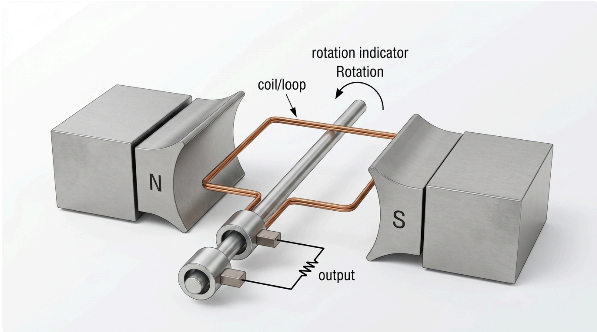
Curriculum context	LO5 — lesson.emf.ac-generation-principles
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a single loop of wire, loop plane facing the poles head-on (widest visible loop face), rotating on a central axis between N and S poles, establishing the physical basis of single-loop AC generation at Level 2 depth.

REFERENCE (AS SENT TO GEMINI)

reference preview not found

Wikimedia Commons — Diagram of single-phase generator with two poles.svg

FINAL ALP ARTWORK (ATTEMPT 1)



Model: gemini-3.1-flash-image · Generated: 2026-08-24T23:21:37.579Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
N/S magnetic poles	PASS	N (left) and S (right) clearly labelled on the pole pieces.
loop/coil between the poles	PASS	Single copper loop visible, positioned between the N and S pole faces.
loop plane facing poles head-on (widest visible loop face) -- defining fact of this asset	PASS	The loop's rectangular face is substantially visible from this viewing angle (not an edge-on sliver), consistent with the near-zero-EMF, face-on pose requested.
central rotational axis	PASS	A single common shaft runs through the loop and both slip rings.
output/slip-ring concept at governed Level-2 abstraction	PASS	TWO separate slip rings are visible, mounted concentrically on the shaft, each contacted by its own stationary brush, with wires running from the brushes to a resistor labelled OUTPUT. This is a materially complete topology, not merely a vague 'concept' -- directly resolves the CC-11.9 handover's complaint about the prior catalogue's under-specified slip-ring wording.
rotating conductor cuts magnetic flux	PASS	Implied correctly by the loop-between-poles + rotation-indicator composition; not contradicted anywhere.

Prohibited change	Result	Note
do not substitute a detailed modern alternator	PASS	Kept at the elementary single-loop/two-slip-ring level matching the DOE reference, not an over-detailed modern machine.

Prohibited change	Result	Note
do not add three-phase windings, phasors or brushes/commutator detail beyond governed scope	PASS	Exactly two brushes, two rings -- no commutator (a commutator would be a single split ring; this is unambiguously two separate full rings), no three-phase winding.
do not depict the edge-on loop pose in this asset -- that is the sibling ProductionAsset	PASS	The loop is shown face-on, not edge-on -- correct pose for the horizontal/near-zero-EMF asset.

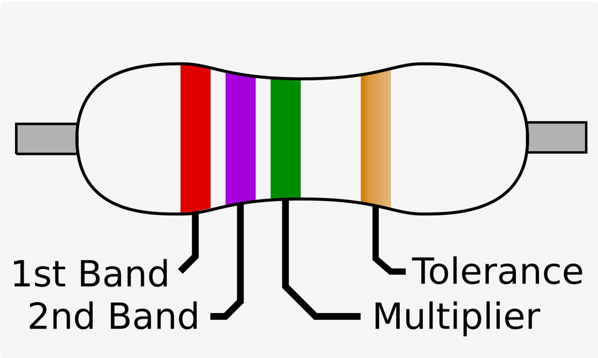
Overall findings	Strong PASS on first attempt. The prepared DOE Figure 1 crop (a genuine, complete, technically authoritative topology reference -- selected via real text search of the handbook for 'slip ring' rather than guessed) gave Gemini an unambiguous topology to redraw, and it preserved every required structural element: two separate slip rings, two brushes, correct output path, no commutator, correct N/S, correct face-on pose. This directly satisfies the CC-11.9 handover's specific concern that the previous generic Commons SVG reference under-specified the slip-ring mechanism.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.generator.rotating-loop.horizontal\unit202.generator.rotating-loop.horizontal-master-v1.png`

Physical electronic component — resistor

unit202.components.physical.resistor

Curriculum context	LO6 — lesson.electrical.electronic-components-passive / -switching-control
Pedagogical role	PHYSICAL_RECOGNITION
Production class	PREMIUM CONCEPTUAL + deterministic UK/IEC symbol
Purpose	A physical-appearance companion image for the resistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

REFERENCE (AS SENT TO GEMINI)



PRIMARY REFERENCE STILL TO BE APPROVED

FINAL ALP ARTWORK (ATTEMPT 1)

Model: gemini-3.1-flash-image · Generated: 2026-08-25T17:50:57.467Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
package form must be a real, representative physical form for a resistor	PASS	A real axial through-hole resistor package: cylindrical body, colour-band sequence (red, violet, green, gold), two wire leads. Matches the reference's own 1st-band/2nd-band/multiplier/tolerance band structure and ordering exactly.

Prohibited change	Result	Note
do not invent a misleading package form for the resistor	PASS	Package form is a standard, recognisable axial resistor -- not misleading.
do not depict any other component in this asset	PASS	Only the resistor is depicted.

Overall findings	Strong PASS on first attempt. Fourth and final four-asset pilot member complete.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.components.physical.resistor\unit202.components.physical.resistor-master-v1.png`